

Introduction to Random Variables

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MATH 241

Week 5, Lecture #1 — Ch. 3.1-3.2, 3.6

Topics for Next Few Weeks

We've worked with methods for finding probabilities, particularly when dealing with finite spaces and a limited number of events. **We will be formalizing these situations into random variables** (denoted as X).

Topics for Next Few Weeks

This will involve exploring the following:

- The use of distributions (e.g. pmfs/pdfs and cumulative density functions)
 - ▶ $f_X(x)$, $F_X(x)$
 - ▶ Defining values of parameters (μ, σ^2, λ)
- Expectation, variance, and moments
 - ▶ $E(X)$, $\text{Var}(X)$, $E(X^2)$
- The use of summations and integrals for finding probabilities, e.g.

$$P(1 \leq x \leq 2) = \int_1^2 2e^{-2x} dx$$

Writing Events Can Become Cumbersome!

- Our partition of events can become expansive and complex
 - ▶ e.g. number of heads possible in 100 coin flips
 - ▶ e.g. chance of a person from a pool finishing first in a marathon
 - ▶ e.g. amount of time it takes for first failure to occur
- The notation becomes cumbersome

Solution: The use of random variables!

Random Variables

Given an experiment with sample space $\mathcal{S}$, a **random variable** X is a function from the sample space $\mathcal{S}$ to the real numbers $\mathbb{R}$. It allows us to map the sample space onto the real line.

- Use of upper-case X , where a numerical value $X(s)$ is assigned to each possible outcome s of the experiment
- Probabilities determined by probability function P , where $P(X(s) = x)$ is an outcome assigned to some x
- We have some support for X , which is all possible values that X can take on
 - ▶ Discrete, continuous, goes from 0 to infinity, etc.

Take-home: X will refer to some random variable and x will be a value of that random variable

Examples: Coin Tosses

Consider an experiment where we roll 2 dice. We know that $\mathcal{S} = \{(1, 1), (1, 2), (1, 3), \dots, (6, 6)\}$.

- We can let random variable X be the sum of the two dice
 - ▶ More formally, $X = X(s) = \text{sum of 2 dice for } s \in \mathcal{S}$

Then what are the realized values of the random variable in the following examples?

Examples: Coin Tosses

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- We can let random variable X be the sum of the two dice
 - ▶ More formally, $X = X(s) = \text{sum of 2 dice for } s \in \mathcal{S}$

Then what are the realized values of the random variable in the following examples?

- $X(\{(1, 1)\}) = 2$
- $X(\{(4, 6)\}) = 10$
- $X(\{(6, 4)\}) = 10$

Writing Random Variables

- X is the random variable, x is a value of that random variable
 - ▶ x can be a result of a number of outcomes from sample space $\mathcal{S}$
- Functions of X
 - ▶ $P(X = x)$, $P(X > x)$, etc.
 - ▶ $f_X(x)$
 - ▶ $E(X)$, $Var(X)$, $E(X^2)$
- Distributed in some way
 - ▶ e.g. $X \sim Bin(n, p)$, $X \sim Norm(0, 1)$

Types of RVs

Random variables can be categorized as one of two types:

① Discrete

- ▶ Random variable takes on counting numbers for support
- ▶ Probability exists at a given value
- ▶ Use of summation for probabilities
- ▶ e.g. Binomial, Poisson, Geometric, Hypergeometric

② Continuous

- ▶ Random variable takes on real space for some support
- ▶ Probability exists between two points
- ▶ Use of integration for probabilities
- ▶ e.g. Normal, Exponential, Uniform, Gamma

Definition 3.2.1: A random variable X is said to be discrete if there is a finite list of values $a_1, a_2, \dots, a_n$ or an infinite list of values $a_1, a_2, \dots$ such that $P(X = a_j \text{ for some } j) = 1$.

- The support of a discrete RV X is the finite or countably infinite set of values of x such that $P(X = x) > 0$
- For a discrete RV, we have a...
 - ▶ Probability mass function (PMF): $p_X(x) = P(X = x)$
 - ▶ Cumulative density function (CDF):
 $F_X(x) = P(X \leq x) = \sum_{a \leq x} P(X = a)$ for all a in support less than x
 - ▶ Expectation: $E(X) = \sum_i x_i f_X(x_i)$
 - ▶ Variance, moments, etc.

A probability mass function (PMF) of a discrete RV X is the function p_X given by:

$$p_X(x) = P(X = x),$$

where x is an element of the support of X .

- We care about probabilities at singular points
- Will be extended to cumulative density functions for discrete RVs
- Its continuous RV version is called a probability density function (PDF)

Example: PMF for Heads in 2 Coin Tosses

Suppose we flip two coins and care about the number of heads appearing on those two coin flips.

- We let X be a random variable corresponding to the number of heads gathered on the two coin flips
 - ▶ What is the support of this random variable? (i.e. the possible number of heads)

Example: PMF for Heads in 2 Coin Tosses

Suppose we flip two coins and care about the number of heads appearing on those two coin flips.

- We let X be a random variable corresponding to the number of heads gathered on the two coin flips
 - ▶ What is the support of this random variable? (i.e. the possible number of heads)
- $P(X = x)$ is determined for each value of x in the support

$$P(X = x) = \begin{cases} \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} & x = 0 \\ 2 \times \frac{1}{2} \times \frac{1}{2} = \frac{2}{4} & x = 1 \\ \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} & x = 2 \end{cases}$$

Probabilities for a PMF can be determined empirically, or through the use of a distribution (as seen later).

Valid PMFs? (3.2.7)

Question: What makes a valid PMF?

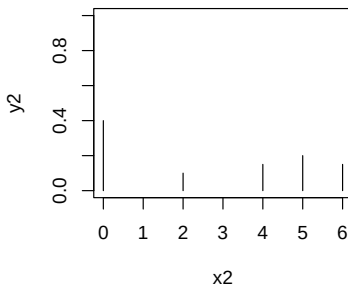
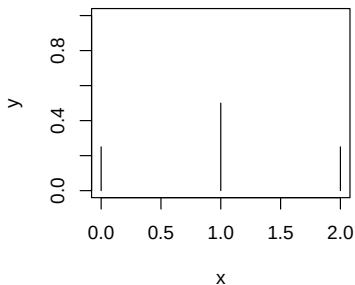
Valid PMFs? (3.2.7)

Question: What makes a valid PMF?

A valid PMF $p_X(x)$ satisfies these two criteria for a discrete RV X with values of the support $x_1, x_2, \dots$:

- **It is nonnegative:** $p_X(x) \geq 0$ for all x
 - ▶ Some probability > 0 for each x in support, 0 otherwise
- **Sums up to 1 across whole support:** $\sum_{i=1}^{\infty} p_X(x_i) = 1$

Visuals



- Visuals of CDFs of discrete RVs

Activity: Discrete RVs

In groups of 2-3: for each of the situations below, write out (1) what an example of a random variable could be and (2) what the PMF is:

- 1 Rolling a single die
- 2 Flipping six coins
- 3 Students within MATH 241

Finding Probabilities of Multiple Values of Support

If we care about a sub-domain of our support and want to find the probability of values in that sub-domain, we can use summation to find the probability:

$$\begin{aligned} P(a \leq X \leq b) &= \sum_{x=a}^b p_X(x) \\ &= \sum_{x=a}^b P(X = x) \end{aligned}$$

- Inclusion matters here!

- ▶ $a < X$ instead of $a \leq X$: start at $a + 1$
- ▶ $X < b$ instead of $X \leq b$: end at $b - 1$

Example: PMF for Heads in 2 Coin Tosses

Suppose we flip two coins and care about getting at least 1 heads in two flips. We have the following PMF for X , representing the number of heads we get in two coin flips:

$$P(X = x) = \begin{cases} \frac{1}{4} & \text{for } x = 0 \\ \frac{1}{2} & \text{for } x = 1 \\ \frac{1}{4} & \text{for } x = 2 \end{cases}$$

Then, for $P(X \geq 1)$:

$$\begin{aligned} P(X \geq 1) &= \sum_{x=1}^2 P(X = x) \\ &= P(X = 1) + P(X = 2) \\ &= \frac{1}{2} + \frac{1}{4} \\ &= \frac{3}{4} \end{aligned}$$

Question: What is another way we could solve for $P(X \geq 1)$?

What we've done motivates our conversation on cumulative density functions $P_X(x)$. A cumulative density function (CDF) of a random variable X is the function $P_X(x)$, given by:

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= \begin{cases} \sum_{a \leq x} P(X = a) & \text{for a discrete RV} \\ \int_{a \leq x} f_X(x) dx & \text{for a continuous RV} \end{cases} \end{aligned}$$

- We will focus on discrete random variables for now, but calculus is definitely needed when looking at CDFs of continuous RVs

Example: CDF for Heads in 2 Coin Tosses

Suppose we flip two coins and care about getting at least 1 heads in two flips. We have the following PMF for X , representing the number of heads we get in two coin flips:

$$P(X = x) = \begin{cases} \frac{1}{4} & \text{for } x = 0 \\ \frac{1}{2} & \text{for } x = 1 \\ \frac{1}{4} & \text{for } x = 2 \end{cases}$$

To find the *CDF* for values of x , consider where we have jumps ($x = 0, x = 1, x = 2$). We will break the CDF into four segments, looking at:

- $x < 0$
- $0 \leq x < 1$
- $1 \leq x < 2$
- $x \geq 2$

Example: CDF for Heads in 2 Coin Tosses

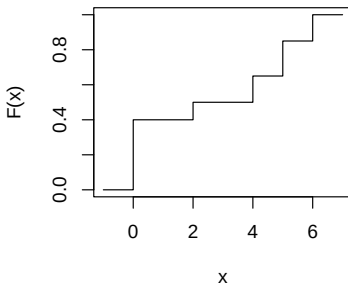
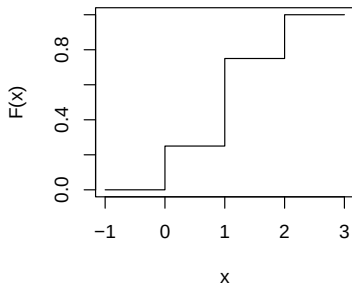
Suppose we flip two coins and care about getting at least 1 heads in two flips. We have the following PMF for X , representing the number of heads we get in two coin flips:

$$P(X = x) = \begin{cases} \frac{1}{4} & \text{for } x = 0 \\ \frac{1}{2} & \text{for } x = 1 \\ \frac{1}{4} & \text{for } x = 2 \end{cases}$$

For each segment, we have the following calculations for the CDF:

$$F_X(x) = \begin{cases} \sum_{a < x} P(X = a) = 0 & \text{for } x < 0 \\ \sum_{a < x} P(X = a) = \frac{1}{4} & \text{for } 0 \leq x < 1 \\ \sum_{a < x} P(X = a) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} & \text{for } 1 \leq x < 2 \\ \sum_{a < x} P(X = a) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1 & \text{for } x \geq 2 \end{cases}$$

Visuals



- Visuals of PMFs of discrete RVs
- Suppose we care about total probability up to some value x ?
 - ▶ More on this in a second

Activity: Discrete RVs

In groups of 2-3: for each of the situations below, write out what the CDF is:

- 1 Rolling a single die
- 2 Flipping six coins
- 3 Students within MATH 241

Valid CDFs? (3.6.7)

Question: What makes a valid CDF?

Question: What makes a valid CDF?

A valid CDF $F_X(x)$ has the following properties:

- **Non-decreasing:** If $x_i \leq x_j$, then $P(X \leq x_i) \leq P(X \leq x_j)$
- **Right-continuous:**

$$F(a) = \lim_{x \rightarrow a^+} F(x)$$

- **Convergence:**

$$\lim_{x \rightarrow -\infty} F(x) = 0$$

$$\lim_{x \rightarrow \infty} F(x) = 1$$

Valid CDFs, A Check (3.6.7)

To verify if a random variable X has a proper CDF, we check the three properties:

- **Non-decreasing:** Is the probability non-decreasing as we traverse left to right from $-\infty$ to ∞ ?
- **Right-continuous:** At each part of the domain, is the inequality left-inclusive? (i.e. $a \leq X$ vs. $a < X$).
- **Convergence:** Substitute $-\infty$ and ∞ into $F_X(x)$. Do they result in 0 or 1?

Activity: Rolling Three Dice

Suppose you roll three dice and are curious about some partition of events. In groups of 2-3,

- 1 Determine a discrete random variable from the situation of rolling 3 dice.
- 2 Calculate the PMF and CDF of this discrete random variable.
- 3 Determine if the PMF and CDF are valid.
- 4 Discuss what patterns your group found in the calculations of the PMFs and CDFs.

Named Distributions

Certain situations have fortunately been found as valid probability distributions. For the discrete RVs, we have the following:

- **Bernoulli**: a single binary outcome, with some parameter p of an outcome occurring
- **Binomial**: n independent, binary trials, with some parameter p of an outcome occurring
 - ▶ **Multinomial**: extension to n independent trials of multiple outcomes with probability $p_1, p_2, \dots, p_n$ occurring
- **Geometric**: first success from binary, independent trials with some parameter p of an outcome occurring
- **Hypergeometric**: k successes from n draws without replacement with K possible successes and N outcomes
- **Discrete uniform**: uniform probability across n values of support, where probability is equal to $1/n$

What's to Come

- These distributions will have the same functional form for their PMFs/PDFs and CDFs
 - ▶ Changes in parameters
 - ▶ Useful for summation and integration techniques
- Working with the PMFs and CDFs
- Use of R to check these results

For Next Week

- Read chapter 3 (sections 3.1-3.7) from Intro to Probability textbook
- No homework due this week, but there will be one for the following week (assigned over the weekend)
- Checkpoint #3 in next Thursday's class
 - ▶ More activity-like than quiz-like; in group settings
- New office hours starting next week:
 - ▶ Monday from 1-2pm
 - ▶ Tuesday from 3-4pm
 - ▶ Friday from 10-11:30am